Assignment - 01

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Section : 09

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Ans. to the ques. no. 1

a Four stages of SDLC -

1) Planning : Why should we build the system?
2) Analysis : Who, What, When, Where will the system be?
3) Design : How will the system work?
4) Implementation : How system delivery be delivered.

b 1) Planning Phase
2) Analysis "
3) Design "
4) Implementation "

c This is agile methodology because it has short iterative sprints, continuous stakeholder's feedback option that is fitting for requirement changes.

Ans. to the ques. no. 2

a) 1) Worked separately where there was communication gaps.

2) Time was wasted which caused delay

b) Eight DevOps steps -

1) Plan
2) Code
3) Build
4) Test
5) Release
6) Deploy
7) Operate
8) Monitor

Ans. to the ques. no. 3

a) Three feasibility character -

1) Technical feasibility : Can we build it?

2) Economic feasibility : Should we build it?

3) Organizational feasibility : If we build it, will they come?

b Technical : Integrating new system with old one can be hard.

Economic : Development cost can be higher and can cross the break even point.

Organizational : Pharmasist might resist the changed workflow.

Ans. to the ques. no. 4

Items	Year 1	Year 2	Year 3	Year 4	Total
Total Benefits	150000	200000	250000	300000	
Development cost	240000	0	0	0	
Operational cost	60000	60000	70000	80000	
Total cost	(240000+60000) = 300000	60000	70000	80000	
PV factor	0.909	0.826	0.751	0.683	
PV of benefits	(150000 x 0.909) = 136350	(200000 x 0.826) = 165200	187750	204900	694200
PV of costs	(300000 x 0.909) = 272700	(60000 x 0.826) = 49560	(70000 x 0.751) = 52570	(80000 x 0.683) = 54640	429470

b Σ PV Benefits = 694200

Σ PV cost = 429470

NPV = (694200 - 429470)

= 264730

c ROI = $\frac{\text{PV Benefits} - \text{PV cost}}{\text{PV cost}}$

= $\frac{694200 - 429470}{429470}$

= 0.6164

= 61.64%.

d

Item	Year 1	Year 2	Year 3	Year 4
Benefits	(136350)	115640	135180	150260
Cumulative Net Cash flow	(136350)	(20710)	114470	264730

$\therefore$ BEP = $2 + \frac{(135180 - 114470)}{135180}$ = 2.15 years

Ans. to the ques. no. 5

Q System Request for online study-room reservation system -

1) Project Sponsor : The Head Librarian is the one who initiates the project and provides resources where needed.

2) Business Need : The library needs a more efficient and reliable process. Room reservation causes double bookings or gives students limited access as it is handled at the front desk manually.

3) Business Requirements : System should let students search availability of rooms by date and time slot. They can also reserve, modify and cancel bookings online.

4) Business Value : Improved student satisfaction and better library reputation.

5) Special Issues : The system should have security and privacy and have authentication process.

b 1) Tangible

2) Intangible

3) Tangible

4) Intangible

Ans. to the ques. no. 6

a A functional requirement states what the system should do, defines all the services required by the customer that must be provided by the system.

A non-functional requirement states how the system will do the tasks and defines its properties and constraints.

b 1) Functional

2) Non-functional : Performance

3) Non-functional : Security

4) Non-functional : Cultural and Political

Ans. to the ques. no. 7

1) Questionnaire : It reaches large group easily and quickly.

2) JAD session : It brings all stakeholders together to resolve conflict

3) Observation : We can see if people really do as they say.

Ans. to the ques. no. 8

a Total processing time = (15+30+20+25+10)
= 100 minutes

Total elapsed time = 6 days (8 hour days)

∴ Total " " = (6 × 8 × 60)
= 2880 minutes

The gap between elapsed time & processed time

$= (2880 - 100)$

$= 2780$ minutes

The gap indicates the waiting or queue time between steps. So its delay dominated.

b) They'll take 30 minutes to complete the task

if verify ~~and~~ document & credit check run

parallely.

$\therefore$ Processing time $= (15 + 30 + 25 + 10)$

$= 80$ minutes

$\therefore$ Saved time $= (100 - 80)$

$= 20$ minutes

c) i) Problem Analysis

ii) Root cause analysis.